10. Find the length of the perpendicular and the co-ordinates of the foot of the perpendicular from $(-10, -2)$ to the line $x + y - 2 = 0$
11. If p_1 and p_2 are the lengths of the perpendiculars from the origin to the straight lines $x \sec\theta + y \operatorname{cosec}\theta = 2a$ and $x\cos\theta - y\sin\theta = a\cos 2\theta$, then prove that $p_1^2 + p_2^2 = a^2$.
12. Find the distance between the parallel lines
    (i) $12x + 5y = 7$ and $12x + 5y + 7 = 0$
    (ii) $3x - 4y + 5 = 0$ and $6x - 8y - 15 = 0$.
13. Find the family of straight lines (i) Perpendicular (ii) Parallel to $3x + 4y - 12 = 0$.
14. If the line joining two points A(2,0) and B(3,1) is rotated about A in anticlockwise direction through an angle of 15°, then find the equation of the line in new position.
15. A ray of light coming from the point (1,2) is reflected at a point A on the x-axis and it passes through the point (5,3). Find the co-ordinates of the point A.
16. A line is drawn perpendicular to $5x = y + 7$. Find the equation of the line if the area of the triangle formed by this line with co-ordinate axes is 10 sq. units.
17. Find the image of the point $(-2, 3)$ about the line $x + 2y - 9 = 0$.
18. A photocopy store charges ₹ 1.50 per copy for the first 10 copies and ₹ 1.00 per copy after the 10th copy. Let x be the number of copies, and let y be the total cost of photocopying. (i) Draw graph of the cost as x goes from 0 to 50 copies. (ii) Find the cost of making 40 copies
19. Find atleast two equations of the straight lines in the family of the lines $y = 5x + b$, for which b and the x-coordinate of the point of intersection of the lines with $3x - 4y = 6$ are integers.
20. Find all the equations of the straight lines in the family of the lines $y = mx - 3$, for which m and the x-coordinate of the point of intersection of the lines with $x - y = 6$ are integers.

6.5 Pair of Straight Lines

The equations of two or more lines can be expressed together by an equation of degree higher than one. As we see that a linear equation in x and y represents a straight line, the product of two linear equations represent two straight lines, that is a pair of straight lines. Hence we study pair of straight lines as a quadratic equations in x and y.

Let $L_1 \equiv a_1x + b_1y + c_1 = 0$ and $L_2 \equiv a_2x + b_2y + c_2 = 0$, be separate equations of two straight lines. If $P(x_1, y_1)$ is a point on L_1, then it satisfies the equaiton $L_1 = 0$. Similarly, if $P(x_1, y_1)$ is on L_2 then $L_2 = 0$. If $P(x_1, y_1)$ lies either on $L_1 = 0$ or $L_2 = 0$, then $P(x_1, y_1)$ satisfies the equation $(L_1)(L_2) = 0$, and no other point satisfies $L_1 \cdot L_2 = 0$. Therefore the equation $L_1 \cdot L_2 = 0$ represents the pair of straight lines $L_1 = 0$ and $L_2 = 0$.

For example, consider the two equations

$$y + \sqrt{3}x = 0 \text{ and } y - \sqrt{3}x = 0.$$

The above two equations represent the equation of two straight lines passing through the origin with slopes $-\sqrt{3}$ and $\sqrt{3}$ respectively.

Combining the above equation, we get

$$\left(y + \sqrt{3}x\right)\left(y - \sqrt{3}x\right) = 0$$

$\Rightarrow y^2 - 3x^2 = 0$, represents the pair of straight lines

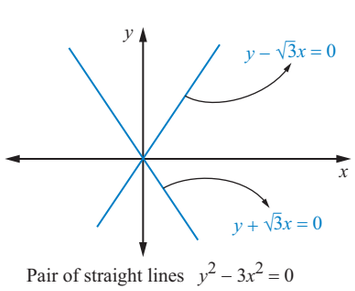


Figure 6.48